King Mongkut's University of Technology Thonburi
Faculty of Engineering, Department of Computer Engineering

Course Syllabus

Course CPE 371 Big Data

Credit Hours 3/0/6 **Class Time** Tue. 09.30-12.20 for International Students
Thu. 13:30-16:20 for HDS Students

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Course Description

The basics of big data analytics and modelling, visualization, structured and unstructured databases, design of experiments, and big data platform with data integration, data processing, workflow management, and data governance. The course will integrate the business applications into every lecture throughout the semester. Professional tools such as Hadoop Ecosystem, Spark and Tableau will be used in practice.

Grading

Midterm examination	30 %
Final examination	30 %
Labs (10 labs)	20 %
Term project	20 %

Course Outline and Schedule

Week	Topic
1	Introduction to Big Data Analytics and Its Framework
2	Hadoop Distributed File System (Lab 1)
3	Hadoop MapReduce (Lab 2)
4	Data Science in Big Data Environments (Lab 3)
5	Data Visualization with Tableau and Hue (Lab 4)
6	Big Data with IoT (Lab 5)
	Midterm Exam
7	Data Service with Hive and Impala (Lab 6)
8	Spark Data Operation (Lab 7)
9	Data Integration Tools with Flume and Sqoop (Lab 8)
10	Database management with HBase (Lab 9)
11	Workflow Management with Oozie and Airflow (Lab 10)
12	Data Governance and Metadata Management
13	Final Project Presentation
	Final examination